Binomial Theorem: Terms and Coefficients — Solutions

Warm-up

Question 1:
Using Pascal's triangle for $n = 4$, the coefficients are $1, 4, 6, 4, 1$.

$$\begin{aligned}(1+x)^4 &= \binom{4}{0}(1)^4(x)^0 + \binom{4}{1}(1)^3(x)^1 + \binom{4}{2}(1)^2(x)^2 + \binom{4}{3}(1)^1(x)^3 + \binom{4}{4}(1)^0(x)^4 \\ &= 1(1) + 4(x) + 6(x^2) + 4(x^3) + 1(x^4) \\ &= 1 + 4x + 6x^2 + 4x^3 + x^4\end{aligned}$$

Question 2:
The general term T_{k+1} of the binomial expansion $(a+b)^n$ is given by the formula:

$$T_{k+1} = \binom{n}{k} a^{n-k} b^k$$

Question 3:
For any binomial expansion $(a+b)^n$, the number of terms in the expanded form is $n+1$.

$$\begin{aligned}\text{Number of terms} &= 12 + 1 \\ &= 13\end{aligned}$$

Question 4:
We expand $(1+2x)^6$ using the binomial theorem:

$$\begin{aligned}T_1 &= \binom{6}{0}(1)^6(2x)^0 = 1 \\ T_2 &= \binom{6}{1}(1)^5(2x)^1 = 6(2x) = 12x \\ T_3 &= \binom{6}{2}(1)^4(2x)^2 = 15(4x^2) = 60x^2\end{aligned}$$

Thus, the first three terms are $1 + 12x + 60x^2$.

Question 5:
The general term of $(1-x)^5$ is $T_{k+1} = \binom{5}{k}(1)^{5-k}(-x)^k$. To find the coefficient of x^2, we set $k = 2$:

$$\begin{aligned}\text{Coefficient} &= \binom{5}{2}(-1)^2 \\ &= \frac{5 \times 4}{2 \times 1}(1) \\ &= 10\end{aligned}$$

Question 6:
The combination formula ${}^nC_r$ (or $\binom{n}{r}$) is defined as:

$$ {}^nC_r = \frac{n!}{r!(n-r)!} $$

Question 7:
In the expansion of $(x+a)^n$, the general term is $T_{k+1} = \binom{n}{k} x^{n-k} a^k$. For a term to be constant, the power of x must be zero:

$$ \begin{aligned} n - k &= 0 \\ \implies k &= n \end{aligned} $$

Thus, the constant term is the $(n+1)$-th term, $T_{n+1} = \binom{n}{n} x^0 a^n = a^n$.

Question 8:
The 4th term corresponds to $k = 3$ in the general term $T_{k+1} = \binom{n}{k} a^{n-k} b^k$. For $(x^2+3)^8$, we have $a = x^2$, $b = 3$, and $n = 8$:

$$ \begin{aligned} T_4 &= \binom{8}{3} (x^2)^{8-3} (3)^3 \\ &= \binom{8}{3} (x^2)^5 (27) \\ &= 56 \cdot 27 x^{10} \end{aligned} $$

The power of x is 10.

Question 9:
The general term of $(2+x)^4$ is $T_{k+1} = \binom{4}{k} (2)^{4-k} (x)^k$. For x^3, we set $k = 3$:

$$ \begin{aligned} \text{Coefficient} &= \binom{4}{3} (2)^{4-3} \\ &= 4 \times 2^1 \\ &= 8 \end{aligned} $$

Question 10:
The expansion of $(x + \frac{1}{x})^2$ is a simple square:

$$ \begin{aligned} \left(x + \frac{1}{x}\right)^2 &= x^2 + 2(x)\left(\frac{1}{x}\right) + \left(\frac{1}{x}\right)^2 \\ &= x^2 + 2 + \frac{1}{x^2} \end{aligned} $$

The term independent of x is 2.

Skill-building

Question 1:
The general term of $(2+x)^7$ is $T_{k+1} = \binom{7}{k}(2)^{7-k}x^k$. For x^3, we set $k=3$:

$$\begin{aligned}\text{Coefficient} &= \binom{7}{3}(2)^{7-3} \\ &= \frac{7\times 6\times 5}{3\times 2\times 1}(2^4) \\ &= 35\times 16 \\ &= 560\end{aligned}$$

Question 2:
The general term of $(x+\frac{1}{x})^4$ is $T_{k+1} = \binom{4}{k}x^{4-k}(x^{-1})^k = \binom{4}{k}x^{4-2k}$. For the term independent of x, we set the exponent to 0:

$$\begin{aligned}4-2k &= 0 \\ \implies 2k &= 4 \\ \implies k &= 2\end{aligned}$$

Substituting $k=2$ into the general term:

$$\begin{aligned}T_3 &= \binom{4}{2}x^{4-4} \\ &= 6\times 1 \\ &= 6\end{aligned}$$

Question 3:
For $(x-3)^8$, there are $8+1=9$ terms. The middle term is the 5th term (T_5), where $k=4$.

$$\begin{aligned}T_5 &= \binom{8}{4}x^{8-4}(-3)^4 \\ &= 70\times x^4\times 81 \\ &= 5670x^4\end{aligned}$$

Question 4:
The general term of $(1+4x)^{10}$ is $T_{k+1} = \binom{10}{k}(1)^{10-k}(4x)^k = \binom{10}{k}4^k x^k$. For x^2, $k=2$: $\text{Coeff}_{x^2} = \binom{10}{2}4^2 = 45\times 16 = 720$. For x^3, $k=3$: $\text{Coeff}_{x^3} = \binom{10}{3}4^3 = 120\times 64 = 7680$.

$$\begin{aligned}\text{Ratio} &= \frac{720}{7680} \\ &= \frac{72}{768} \\ &= \frac{3}{32}\end{aligned}$$

Question 5:
The general term of $(2x-3)^9$ is $T_{k+1} = \binom{9}{k}(2x)^{9-k}(-3)^k$. For x^5, we need $9-k=5$, so $k=4$:

$$\begin{aligned}\text{Coefficient} &= \binom{9}{4}(2)^{9-4}(-3)^4 \\ &= 126 \times 2^5 \times 81 \\ &= 126 \times 32 \times 81 \\ &= 326592\end{aligned}$$

Question 6:
The general term of $(1+kx)^5$ is $T_{k'+1} = \binom{5}{k'}(1)^{5-k'}(kx)^{k'}$. For x^2, $k'=2$:

$$\begin{aligned}\text{Coefficient} &= \binom{5}{2}k^2 \\ &= 10k^2\end{aligned}$$

Given that the coefficient is 40:

$$\begin{aligned}10k^2 &= 40 \\ \implies k^2 &= 4 \\ \implies k &= \pm 2\end{aligned}$$

Question 7:
The general term of $(3x^2 - \frac{1}{x})^5$ is $T_{k+1} = \binom{5}{k}(3x^2)^{5-k}(-x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned}T_{k+1} &= \binom{5}{k}3^{5-k}(x^2)^{5-k}(-1)^k(x^{-1})^k \\ &= \binom{5}{k}3^{5-k}(-1)^k x^{10-2k-k} \\ &= \binom{5}{k}3^{5-k}(-1)^k x^{10-3k}\end{aligned}$$

For x^4, we set $10-3k=4$:

$$\begin{aligned}\implies 3k &= 6 \\ \implies k &= 2\end{aligned}$$

Substituting $k=2$:

$$\begin{aligned}T_3 &= \binom{5}{2}3^{5-2}(-1)^2 x^{10-6} \\ &= 10 \times 3^3 \times 1 \times x^4 \\ &= 270x^4\end{aligned}$$

Question 8:
Using the factorial definition of combinations:

$$\begin{aligned} {}^nC_r &= \frac{n!}{r!(n-r)!} \\ \text{And } {}^nC_{n-r} &= \frac{n!}{(n-r)!(n-(n-r))!} \\ &= \frac{n!}{(n-r)!r!} \end{aligned}$$

Since both expressions are identical, ${}^nC_r = {}^nC_{n-r}$.

Question 9:
We need the coefficient of x^3 in $(1+x)^2(1-x)^6$. $(1+x)^2 = 1+2x+x^2$. $(1-x)^6 = \binom{6}{0} - \binom{6}{1}x + \binom{6}{2}x^2 - \binom{6}{3}x^3 + \dots = 1 - 6x + 15x^2 - 20x^3 + \dots$ To get x^3, we multiply terms from the first bracket by terms from the second:

$$\begin{aligned} \text{Coeff}_{x^3} &= (1)(-20) + (2x)(15x^2) + (x^2)(-6x) \\ &= -20 + 30 - 6 \\ &= 4 \end{aligned}$$

Question 10:
The general term of $(2x^3 - \frac{1}{x^3})^4$ is $T_{k+1} = \binom{4}{k}(2x^3)^{4-k}(-x^{-3})^k$. Simplify the powers of x:

$$\begin{aligned} T_{k+1} &= \binom{4}{k}2^{4-k}(x^3)^{4-k}(-1)^k(x^{-3})^k \\ &= \binom{4}{k}2^{4-k}(-1)^k x^{12-3k-3k} \\ &= \binom{4}{k}2^{4-k}(-1)^k x^{12-6k} \end{aligned}$$

For the term independent of x, set $12-6k=0$:

$$\begin{aligned} &\implies 6k = 12 \\ &\implies k = 2 \end{aligned}$$

Substituting $k=2$:

$$\begin{aligned} T_3 &= \binom{4}{2}2^{4-2}(-1)^2 \\ &= 6 \times 4 \times 1 \\ &= 24 \end{aligned}$$

Easier Exam Questions

Question 1:
The general term of $(2x-1)^{10}$ is $T_{k+1} = \binom{10}{k}(2x)^{10-k}(-1)^k$. For x^4, we set $10-k=4$, so $k=6$:

$$\begin{aligned}
\text{Coefficient} &= \binom{10}{6}(2)^{10-6}(-1)^6 \\
&= \binom{10}{4}(2^4)(1) \\
&= 210 \times 16 \\
&= 3360
\end{aligned}$$

Correct option is (D).

Question 2:
Analyze the statements for $(1+x)^7$: (A) Term $T_{k+1} = \binom{7}{k}x^k$. For $k=5$, $T_6 = \binom{7}{5}x^5 = 21x^5$. (True) (B) Constant term is $T_1 = \binom{7}{0}x^0 = 1$. (True) (C) General term is $\binom{7}{k}x^k$. (True) (D)

Number of terms is $n+1 = 7+1 = 8$. (False) Therefore, statement (D) is not true.

Question 3:
The general term of $(5x^2 - \frac{3}{x})^6$ is $T_{k+1} = \binom{6}{k}(5x^2)^{6-k}(-3x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned}
T_{k+1} &= \binom{6}{k}5^{6-k}(x^2)^{6-k}(-3)^k(x^{-1})^k \\
&= \binom{6}{k}5^{6-k}(-3)^k x^{12-2k-k} \\
&= \binom{6}{k}5^{6-k}(-3)^k x^{12-3k}
\end{aligned}$$

For x^6, set $12-3k=6$:

$$\begin{aligned}
&\implies 3k = 6 \\
&\implies k = 2
\end{aligned}$$

Substituting $k=2$:

$$\text{Coefficient} = \binom{6}{2}5^{6-2}(-3)^2 = \binom{6}{4}5^4 3^2$$

Correct option is (D).

Question 4:
The general term of $(2x - \frac{1}{x^2})^8$ is $T_{k+1} = \binom{8}{k}(2x)^{8-k}(-x^{-2})^k$. Simplify the powers of x:

$$
\begin{aligned}
T_{k+1} &= \binom{8}{k} 2^{8-k} x^{8-k} (-1)^k x^{-2k} \\
&= \binom{8}{k} 2^{8-k} (-1)^k x^{8-3k}
\end{aligned}
$$

For x^5, set $8 - 3k = 5$:

$$
\begin{aligned}
&\implies 3k = 3 \\
&\implies k = 1
\end{aligned}
$$

Substituting $k = 1$:

$$
\begin{aligned}
T_2 &= \binom{8}{1} 2^{8-1} (-1)^1 x^{8-3} \\
&= 8 \times 128 \times (-1) x^5 \\
&= -1024 x^5
\end{aligned}
$$

Question 5:
The 6th term corresponds to $k = 5$ in the general term $T_{k+1} = \binom{n}{k} a^{n-k} b^k$. For $(2x+3)^8$, $a = 2x$, $b = 3$, $n = 8$:

$$
\begin{aligned}
T_6 &= \binom{8}{5} (2x)^{8-5} (3)^5 \\
&= \binom{8}{5} (2x)^3 (3)^5
\end{aligned}
$$

Correct option is (D).

Question 6:
The general term of $(4x^3 + \frac{3}{x})^8$ is $T_{k+1} = \binom{8}{k} (4x^3)^{8-k} (3x^{-1})^k$. Simplify the powers of x:

$$
\begin{aligned}
T_{k+1} &= \binom{8}{k} 4^{8-k} (x^3)^{8-k} 3^k x^{-k} \\
&= \binom{8}{k} 4^{8-k} 3^k x^{24-3k-k} \\
&= \binom{8}{k} 4^{8-k} 3^k x^{24-4k}
\end{aligned}
$$

For the constant term, set $24 - 4k = 0$:

$$
\begin{aligned}
&\implies 4k = 24 \\
&\implies k = 6
\end{aligned}
$$

Substituting $k = 6$:

$$
\begin{aligned}
T_7 &= \binom{8}{6} 4^{8-6} 3^6 \\
&= 28 \times 16 \times 729 \\
&= 326592
\end{aligned}
$$

Question 7:
The general term of $(x + \frac{4}{x})^5$ is $T_{k+1} = \binom{5}{k} x^{5-k}(4x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned} T_{k+1} &= \binom{5}{k} x^{5-k} 4^k x^{-k} \\ &= \binom{5}{k} 4^k x^{5-2k} \end{aligned}$$

For the coefficient of x, set $5 - 2k = 1$:

$$\begin{aligned} \implies 2k &= 4 \\ \implies k &= 2 \end{aligned}$$

Substituting $k = 2$:

$$\text{Coefficient} = \binom{5}{2} 4^2 = 10 \times 16 = 160$$

Question 8:
The general term of $(x^2 + \frac{2}{x})^{15}$ is $T_{k+1} = \binom{15}{k}(x^2)^{15-k}(2x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned} T_{k+1} &= \binom{15}{k} x^{30-2k} 2^k x^{-k} \\ &= \binom{15}{k} 2^k x^{30-3k} \end{aligned}$$

For the term independent of x, set $30 - 3k = 0$:

$$\begin{aligned} \implies 3k &= 30 \\ \implies k &= 10 \end{aligned}$$

Substituting $k = 10$:

$$T_{11} = \binom{15}{10} 2^{10} = 3003 \times 1024 = 3075072$$

Question 9:
The general term of $(x^2 + \frac{2}{x})^7$ is $T_{k+1} = \binom{7}{k}(x^2)^{7-k}(2x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned} T_{k+1} &= \binom{7}{k} x^{14-2k} 2^k x^{-k} \\ &= \binom{7}{k} 2^k x^{14-3k} \end{aligned}$$

For the coefficient of x^2, set $14 - 3k = 2$:

$$\implies 3k = 12$$
$$\implies k = 4$$

Substituting $k = 4$:

$$\text{Coefficient} = \binom{7}{4}2^4 = 35 \times 16 = 560$$

Question 10a:
We use the factorial definition of combinations:

$$\begin{aligned}
\text{RHS} &= \binom{n-1}{k-1} + \binom{n-1}{k} \\
&= \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-1-k)!} \\
&= \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-k-1)!}
\end{aligned}$$

To add these, find a common denominator $k!(n-k)!$:

$$\begin{aligned}
\text{RHS} &= \frac{(n-1)!k}{k!(n-k)!} + \frac{(n-1)!(n-k)}{k!(n-k)!} \\
&= \frac{(n-1)![k+(n-k)]}{k!(n-k)!} \\
&= \frac{(n-1)!n}{k!(n-k)!} \\
&= \frac{n!}{k!(n-k)!} = \binom{n}{k}
\end{aligned}$$

Thus, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Question 10b:
The general term of $(2x^2 - \frac{1}{x})^{12}$ is $T_{k+1} = \binom{12}{k}(2x^2)^{12-k}(-x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned}
T_{k+1} &= \binom{12}{k}2^{12-k}x^{24-2k}(-1)^k x^{-k} \\
&= \binom{12}{k}2^{12-k}(-1)^k x^{24-3k}
\end{aligned}$$

For the term independent of x, set $24 - 3k = 0$:

$$\implies 3k = 24$$
$$\implies k = 8$$

Substituting $k = 8$:

$$T_9 = \binom{12}{8}2^{12-8}(-1)^8 = 495 \times 16 \times 1 = 7920$$

Question 11:
The general term of $(2x - \frac{1}{x^2})^9$ is $T_{k+1} = \binom{9}{k}(2x)^{9-k}(-x^{-2})^k$. Simplify the powers of x:

$$\begin{aligned} T_{k+1} &= \binom{9}{k} 2^{9-k} x^{9-k} (-1)^k x^{-2k} \\ &= \binom{9}{k} 2^{9-k} (-1)^k x^{9-3k} \end{aligned}$$

For the term independent of x, set $9 - 3k = 0$:

$$\begin{aligned} \implies 3k &= 9 \\ \implies k &= 3 \end{aligned}$$

Substituting $k = 3$:

$$T_4 = \binom{9}{3} 2^{9-3} (-1)^3 = 84 \times 64 \times (-1) = -5376$$

Question 12:
The general term of $(2x - \frac{1}{x})^{11}$ is $T_{k+1} = \binom{11}{k}(2x)^{11-k}(-x^{-1})^k$. Simplify the powers of x:

$$\begin{aligned} T_{k+1} &= \binom{11}{k} 2^{11-k} x^{11-k} (-1)^k x^{-k} \\ &= \binom{11}{k} 2^{11-k} (-1)^k x^{11-2k} \end{aligned}$$

For the coefficient of x^3, set $11 - 2k = 3$:

$$\begin{aligned} \implies 2k &= 8 \\ \implies k &= 4 \end{aligned}$$

Substituting $k = 4$:

$$\text{Coefficient} = \binom{11}{4} 2^{11-4} (-1)^4 = 330 \times 128 \times 1 = 42240$$

Harder Exam Questions

Question 1:
Given $(1 + kx)^n = 1 - 12x + 60x^2 - \dots$ Using the binomial expansion:

$$\begin{aligned} (1 + kx)^n &= \binom{n}{0}(1)^n + \binom{n}{1}(1)^{n-1}(kx)^1 + \binom{n}{2}(1)^{n-2}(kx)^2 + \dots \\ &= 1 + nkx + \frac{n(n-1)}{2}(kx)^2 + \dots \end{aligned}$$

Comparing coefficients of x:

$$nk = -12$$
$$\implies k = -\frac{12}{n}$$

Comparing coefficients of x^2:

$$\frac{n(n-1)}{2}k^2 = 60$$

Substitute $k = -\frac{12}{n}$ into the second equation:

$$\begin{aligned}
\frac{n(n-1)}{2}\left(-\frac{12}{n}\right)^2 &= 60\\
\frac{n(n-1)}{2}\frac{144}{n^2} &= 60\\
\frac{72(n-1)}{n} &= 60\\
\implies 72n - 72 &= 60n\\
\implies 12n &= 72\\
\implies n &= 6
\end{aligned}$$

Now find k:

$$k = -\frac{12}{6} = -2$$

Thus, $n = 6$ and $k = -2$.

Question 2:
The coefficient of x^n in $(1+x)^n$ is $\binom{n}{n} = 1$. The coefficient of x^n in $(1-x)^n$ is $\binom{n}{n}(-1)^n = (-1)^n$. Summing these coefficients:

$$\text{Total Coefficient} = 1 + (-1)^n$$

Correct option is (C).

Question 3:
First, simplify the ratio:

$$\begin{aligned}
\frac{{}^{10}C_k}{{}^{9}C_k} &= \frac{\frac{10!}{k!(10-k)!}}{\frac{9!}{k!(9-k)!}}\\
&= \frac{10!}{k!(10-k)!} \times \frac{k!(9-k)!}{9!}\\
&= \frac{10 \times 9!}{9!} \times \frac{(9-k)!}{(10-k)(9-k)!}\\
&= \frac{10}{10-k}
\end{aligned}$$

(i) Expand $(x+\frac{2}{x})^5$:

$$\begin{aligned}(x+\frac{2}{x})^5 &= \binom{5}{0}x^5+\binom{5}{1}x^4(\frac{2}{x})+\binom{5}{2}x^3(\frac{2}{x})^2+\binom{5}{3}x^2(\frac{2}{x})^3+\binom{5}{4}x(\frac{2}{x})^4+\binom{5}{5}(\frac{2}{x})^5\\ &= x^5+5x^4(\frac{2}{x})+10x^3(\frac{4}{x^2})+10x^2(\frac{8}{x^3})+5x(\frac{16}{x^4})+\frac{32}{x^5}\\ &= x^5+10x^3+40x+\frac{80}{x}+\frac{80}{x^3}+\frac{32}{x^5}\end{aligned}$$

(ii) Simplify $(x+\frac{2}{x})^5+(x-\frac{2}{x})^5$: Let $f(x)=(x+\frac{2}{x})^5$. The expansion of $(x-\frac{2}{x})^5$ will have the same coefficients but alternating signs for terms with odd powers of $(\frac{2}{x})$:

$$(x-\frac{2}{x})^5 = x^5-10x^3+40x-\frac{80}{x}+\frac{80}{x^3}-\frac{32}{x^5}$$

Summing them:

$$\begin{aligned}\text{Sum} &= (x^5+10x^3+40x+\frac{80}{x}+\frac{80}{x^3}+\frac{32}{x^5})+(x^5-10x^3+40x-\frac{80}{x}+\frac{80}{x^3}-\frac{32}{x^5})\\ &= 2(x^5+40x+\frac{80}{x^3})\\ &= 2x^5+80x+\frac{160}{x^3}\end{aligned}$$

Question 4:
First, simplify the ratio:

$$\frac{^{10}C_k}{^{9}C_k} = \frac{10}{10-k} \text{ (as calculated in Q3)}$$

Now consider $(\frac{x^3}{2}+\frac{a}{x})^8$. The general term is $T_{k+1}=\binom{8}{k}(\frac{x^3}{2})^{8-k}(ax^{-1})^k$. Simplify the powers of x:

$$\begin{aligned}T_{k+1} &= \binom{8}{k}(\frac{1}{2})^{8-k}x^{24-3k}a^kx^{-k}\\ &= \binom{8}{k}(\frac{1}{2})^{8-k}a^kx^{24-4k}\end{aligned}$$

For the constant term, set $24-4k=0$:

$$\begin{aligned}&\implies 4k=24\\ &\implies k=6\end{aligned}$$

Substituting $k=6$:

$$\begin{aligned}\text{Constant term} &= \binom{8}{6}(\frac{1}{2})^{8-6}a^6\\ &= 28\times\frac{1}{4}\times a^6\\ &= 7a^6\end{aligned}$$

Given the constant term is 5103:

$$\begin{aligned} 7a^6 &= 5103 \\ \implies a^6 &= 729 \\ \implies a &= \pm\sqrt[6]{729} \\ \implies a &= \pm 3 \end{aligned}$$

Question 5:
We need the coefficient of x^3 in $(1+x+x^2)^5$. This is a trinomial expansion. The general term is $\frac{5!}{a!b!c!}(1)^a(x)^b(x^2)^c$ where $a+b+c=5$ and the power of x is $b+2c=3$. Possible values for (a,b,c): 1. If $c=1$, then $b=1$. Since $a+b+c=5$, $a=3$. $(3,1,1)$. 2. If $c=0$, then $b=3$. Since $a+b+c=5$, $a=2$. $(2,3,0)$. Calculate the coefficients for each case:

$$\begin{aligned} \text{Case 1: } \frac{5!}{3!1!1!} &= \frac{120}{6} = 20 \\ \text{Case 2: } \frac{5!}{2!3!0!} &= \frac{120}{2\times 6} = 10 \end{aligned}$$

Summing the coefficients:

$$\text{Total Coefficient} = 20 + 10 = 30$$

Correct option is (C).

Question 6:
The general term of $(ax+b)^6$ is $T_{k+1} = \binom{6}{k}(ax)^{6-k}b^k$. For x^4, $6-k=4 \implies k=2$:

$$\begin{aligned} \text{Coeff}_{x^4} &= \binom{6}{2}a^4b^2 = 15a^4b^2 = 2160 \\ \implies a^4b^2 &= 144 \end{aligned}$$

For x^5, $6-k=5 \implies k=1$:

$$\begin{aligned} \text{Coeff}_{x^5} &= \binom{6}{1}a^5b^1 = 6a^5b = -576 \\ \implies a^5b &= -96 \end{aligned}$$

Divide the two equations:

$$\begin{aligned} \frac{a^4b^2}{a^5b} &= \frac{144}{-96} \\ \frac{b}{a} &= -\frac{3}{2} \\ \implies b &= -\frac{3}{2}a \end{aligned}$$

Substitute b back into $a^5b = -96$:

$$
\begin{aligned}
a^5(-\frac{3}{2}a) &= -96 \\
-\frac{3}{2}a^6 &= -96 \\
\implies a^6 &= 64 \\
\implies a &= \pm 2
\end{aligned}
$$

If $a = 2$, then $b = -\frac{3}{2}(2) = -3$. If $a = -2$, then $b = -\frac{3}{2}(-2) = 3$. Thus, $(a, b) = (2, -3)$ or $(-2, 3)$.

Question 7:
(i) The general term of $(1 + 2x)^n$ is $T_{k+1} = \binom{n}{k}(1)^{n-k}(2x)^k = \binom{n}{k}2^k x^k$. For x^4, $k = 4$:

$$\text{Coefficient} = \binom{n}{4}2^4 = 16\binom{n}{4}$$

(ii) For x^6, $k = 6$:

$$\text{Coefficient} = \binom{n}{6}2^6 = 64\binom{n}{6}$$

Given the ratio is 5 : 8:

$$
\begin{aligned}
\frac{16\binom{n}{4}}{64\binom{n}{6}} &= \frac{5}{8} \\
\frac{\binom{n}{4}}{4\binom{n}{6}} &= \frac{5}{8} \\
\implies \binom{n}{4} &= \frac{20}{8}\binom{n}{6} = \frac{5}{2}\binom{n}{6}
\end{aligned}
$$

Expand the combinations:

$$
\begin{aligned}
\frac{n!}{4!(n-4)!} &= \frac{5}{2}\frac{n!}{6!(n-6)!} \\
\frac{1}{24(n-4)(n-5)(n-6)!} &= \frac{5}{2}\frac{1}{720(n-6)!} \\
\frac{1}{24(n-4)(n-5)} &= \frac{5}{1440} \\
\frac{1}{(n-4)(n-5)} &= \frac{5 \times 24}{1440} = \frac{120}{1440} = \frac{1}{12} \\
\implies (n-4)(n-5) &= 12 \\
\implies n^2 - 9n + 20 &= 12 \\
\implies n^2 - 9n + 8 &= 0 \\
\implies (n-8)(n-1) &= 0
\end{aligned}
$$

Since $n \geq 6$, $n = 8$.

Question 8:
The general term of $(1 + \frac{x}{k})^n$ is $T_{r+1} = \binom{n}{r}(1)^{n-r}(\frac{x}{k})^r = \binom{n}{r}k^{-r}x^r$. Coefficient of x^2 ($r = 2$):

$$C_2 = \binom{n}{2} k^{-2} = \frac{n(n-1)}{2k^2}$$

Coefficient of x^3 $(r = 3)$:

$$C_3 = \binom{n}{3} k^{-3} = \frac{n(n-1)(n-2)}{6k^3}$$

Given $C_3 = 2C_2$:

$$\begin{aligned} \frac{n(n-1)(n-2)}{6k^3} &= 2\frac{n(n-1)}{2k^2} \\ \frac{n-2}{6k^3} &= \frac{1}{k^2} \text{ (dividing by } n(n-1) \text{ since } n > 2) \end{aligned}$$

Multiply by $6k^3$:

$$\begin{aligned} n - 2 &= 6k \\ \implies n &= 6k + 2 \end{aligned}$$

Thus, $n = 6k + 2$ is proven.

Question 9:
The general term of $(5x+2)^{20}$ is $T_{k+1} = \binom{20}{k}(5x)^{20-k}2^k$. The coefficient of x^{20-k} is $\binom{20}{k}5^{20-k}2^k$. Let $m = 20 - k$ be the power of x. Then the coefficient of x^m is $\binom{20}{m}5^m 2^{20-m}$. We are given that the coefficients of x^k and x^{k+1} are equal:

$$\begin{aligned} \binom{20}{k} 5^k 2^{20-k} &= \binom{20}{k+1} 5^{k+1} 2^{20-(k+1)} \\ \frac{20!}{k!(20-k)!} 5^k 2^{20-k} &= \frac{20!}{(k+1)!(19-k)!} 5^{k+1} 2^{19-k} \end{aligned}$$

Divide both sides by $20!$, 5^k, and 2^{19-k}:

$$\begin{aligned} \frac{2^1}{k!(20-k)!} &= \frac{5^1}{(k+1)!(19-k)!} \\ \frac{2}{k!(20-k)(19-k)!} &= \frac{5}{(k+1)k!(19-k)!} \\ \frac{2}{20-k} &= \frac{5}{k+1} \end{aligned}$$

Cross-multiply:

$$\begin{aligned} 2(k+1) &= 5(20-k) \\ \implies 2k + 2 &= 100 - 5k \\ \implies 7k &= 98 \\ \implies k &= 14 \end{aligned}$$